

**THERMAL ENGINEERING - I**

Time Allowed: 3 Hours

Full Marks: 70

Answer to Question No.1 is compulsory and to be answered first.

This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided.

On early submission of answer scripts of Question No.1, a student will get the remaining script earlier.

Answer any five questions from Group-A & B, as directed.

1. Answer the following questions (any twenty):

20x1

Choose the correct alternatives from the following:

- i) A closed system receives 60 kJ of heat but its internal energy decreases by 30 kJ. Then the work done by the system is – (a) -90 kJ, (b) -30 kJ, (c) 30 kJ, (d) 90 kJ.
- ii) The internal energy ( $u$ ) of a dry steam is given by, [ $p$  is in bar,  $v_g$  is in  $\text{m}^3/\text{kg}$ ,  $u$  is in  $\text{kJ/kg}$ ] – (a)  $u = h_g + 100 p v_g$  (b)  $u = h_g - 100 p v_g$ , (c)  $u = h_g / 100 p v_g$  (d)  $u = 100 p v_g$ .
- iii) Which of the following is an intensive property of a thermodynamic system? – (a) volume, (b) temperature, (c) energy, (d) mass.
- iv) A composite wall of a furnace has 3 layers of equal thickness having thermal conductivities in the ratio of 1:2:4. What will be the temperature drop ratio across the three respective layer? – (a) 1:2:4, (b) 4:2:1, (c) 1:1:1, (d)  $\log 4: \log 2: \log 1$ .
- v) Entropy may be expressed as a function of – (a) pressure and temperature, (b) temperature and volume, (c) heat and work, (d) both (a) & (b).
- vi) The value of universal gas constant ( $R_u$ ) is – (a) 83.14 J/kgK, (b) 8.314 J/kgK, (c) 831.4 J/kgK, (d) 8314 J/kgK.
- vii) In 100 kg wet saturated steam, mass of the water particle is 10 kg. The dryness fraction of the steam will be – (a) 0.1, (b) 1, (c) 0.85, (d) 0.9.
- viii) The change in internal energy is zero during – (a) adiabatic process, (b) polytropic process, (c) isothermal process, (d) throttling.
- ix) When gas is heated at constant pressure – (a) raises the temperature of the gas, (b) increases the internal energy of the gas, (c) does some external work during expansion, (d) both (b) & (c), (e) both (a) & (b).
- x) The general law of expansion or compression is  $PV^n = \text{constant}$ . The process is said to be hyperbolic if  $n$  is equal to – (a) 0, (b)  $\gamma$ , (c) 1, (d)  $\infty$ .
- xi) If a heat engine gives an output of 3 kW when the input is 10000 J/sec, then the thermal efficiency of the engine will be \_\_\_\_\_.
- xii) The specific volume of water \_\_\_\_\_ when heated from  $0^\circ\text{C}$ . (decreases/ /increases/ first decreases, then increases / first increases, then decreases)

- xiii) The emissivity is highest for – (a) Rubber, (b) Paper, (c) iron oxide, (d) Carbon.
- xiv) The highest value of thermal conductivity is expected for – (a) Water, (b) Solid ice, (c) Melting ice, (d) Steam.

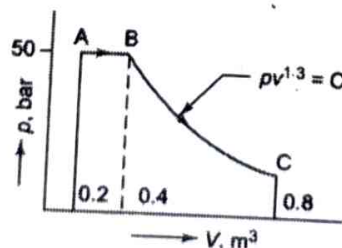
State whether the following statements are True or False:

- xv) The perpetual motion machine of first kind is possible.
- xvi) Zeroth law of thermodynamics deals with temperature.
- xvii) Ratio of two specific heats of a perfect gas is always varies with temperature.
- xviii) In isothermal process change in internal energy is zero.
- xix) Efficiency of any heat engine cycle is always 100%.
- xx) Entropy of universe is increasing.
- xxi) Quasistatic process is a reversible process.
- xxii) Heat is low grade energy.

### Group-A

Answer any three questions.

2. a) Write down the advantages and disadvantages of Geothermal Energy.  
b) Write short notes on: - Solar water heater, Biogas plant. 4+6
3. a) Differentiate between extensive and intensive properties with example.  
b) Prove that heat transfer is a path function.  
c) Show that the work done for an isothermal process from the state 1 to state 2 is given by  $W_{1-2} = p_1 v_1 (\ln p_1 - \ln p_2)$ , where each term has its usual meaning. 3+3+4
4. a) What is steady flow process? Write the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms of it.  
b) A gaseous system undergoes three quasi-static processes in sequence. The gas initially at 5 bar and  $0.01 \text{ m}^3$  is expanded at constant pressure. It is then further expanded according to the relationship  $p v^2 = \text{constant}$  to 2 bar and  $0.025 \text{ m}^3$ . The gas is then returned to its initial state during which  $p v = \text{constant}$ . Sketch the cycle in the P-V diagram. Calculate the work done in each case. (1+4)+(1+4)
5. a) Determine the total work done by a gas system following an expansion process as shown in Figure.



- b) A mass of 8 kg gas expands within a flexible container so that the p-v relationship is of the form  $p v^{1.2} = \text{constant}$ . The initial pressure is 1000 kPa and the initial volume is  $1 \text{ m}^3$ . The final pressure is 5 kPa. If the specific internal energy of the gas decreases by 40 KJ/Kg, find the heat transfer in magnitude and direction. 5+5
6. a) Prove that for an ideal gas, change in entropy,  $S_2 - S_1 = c_v \ln \left( \frac{T_2}{T_1} \right) + R \ln \left( \frac{v_2}{v_1} \right)$ .  
b) An inventor claims to have heat engine which is capable of developing 9 kW while working between the temperature limits of  $30^\circ\text{C}$  and  $-40^\circ\text{C}$ . It receives only 1000 kJ/min of heat. Discuss the possibility of the claim. 5+5

7. a) Establish the relation:  $(C.O.P.)_{HP} = (C.O.P.)_R + 1$   
 b) A mass of 50 kg/min enters the control volume of a steady flow system at 2 bar and 100°C and an elevation of 100 m above the datum. The same mass leaves the control volume at 150 m elevation with a pressure of 10 bar and temperature of 300°C. The entrance and exit velocities are 2400 m/min and 1200 m/min respectively. During the process, 50,000 kJ/hr of heat are transferred to the control volume and the rise in the enthalpy is 8 kJ/kg. Calculate the power developed. 3+7
8. a) State Kelvin-Planck and Celsius statements related to 2<sup>nd</sup> law of thermodynamics.  
 b) 10 kg of air is compressed adiabatically from 1 bar pressure and 18°C to 7 bar and then cooled at constant pressure to original temperature. The air is then expanded adiabatically to the original pressure. [Take  $\gamma = 1.4$  and  $C_p = 1.005 \text{ kJ/kgK}$ ]. Find – (i) The temperature after compression, (ii) The final temperature and (iii) Heat extracted during constant pressure cooling. 4+6

### Group-B

Answer any two questions.

9. a) Describe the working principle of Throttling calorimeter with neat sketch.  
 b) A rigid closed tank of volume 3 m<sup>3</sup> contains 5 kg of wet steam at a pressure of 200 kPa. The tank is heated until the steam becomes dry saturated. Determine the final pressure and the heat transfer to the tank. 4+6
10. a) Define dryness fraction of steam. What do you mean by degree of superheat?  
 b) Write down the advantages of using Superheated steam.  
 c) In a throttling calorimeter, the steam is admitted at a pressure of 10 bar. If it is discharged at atmospheric pressure and 110°C after throttling, determine the dryness fraction of steam. Assume specific heat as steam as 2.2 kJ/kg K. 4+3+3
11. a) State Fourier's law of heat conduction.  
 b) A composite wall consisting of three different layers in perfect thermal contact is shown in sketch below. The outer surfaces to the extreme left and right are kept at temperatures 400°C and 50°C respectively. The thickness  $L_i$  for  $i = 1, 2, 3$  of each layer are given as 5cm, 10cm and 15 cm and the thermal conductivity  $k_i$  for  $i = 1, 2, 3$  of each layer are given as 20W/m°C, 50W/m°C and 100W/m°C respectively. Determine the heat transfer rate per square meter across this composite layer by assuming one dimensional heat flow and using the thermal resistance concept. 3+7
12. a) Distinguish between – (i) Black body and gray body, (ii) Absorptivity and emissivity of a surface.  
 b) Deduce the formula of overall heat transfer co-efficient for a flat composite wall, taking into consideration the convective heat transfer. 4+6